# LESSON PLAN

## HELICOPTER THEORY (11)

| | | | |
|---|---|---|---|
| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 04 |
| **Total No. of Lesson Plans** | 27 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

| | |
|---|---|
| **1. Topic** | Definition of streamline, stream tube, stream filament, ideal and real fluid, steady and turbulent flow, compressible and incompressible flow |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Definition of streamline, stream tube, stream filament, ideal and real fluid, steady and turbulent flow, compressible and incompressible flow, viscous and in viscous flow, rotational and irrotational flow. |
| **5. Specific Objectives** | 11.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

| | |
|---|---|
| **7. Practical in Lab/Equipment Rooms/Simulators** | Review of flow diagrams from IB; Instructor supervision required for diagram interpretation. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

| | |
|---|---|
| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What is the basic definition of a fluid?
(b) Can you name two properties of the atmosphere?
(c) What is the difference between a liquid and a gas in terms of flow?

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Definition of streamline<br>• Concept of stream filament<br>• Definition of stream tube<br>• Properties of ideal fluid<br>• Properties of real fluid | L & Ds | PPT, IB | 15 min |
| (b) | • Steady and turbulent flow<br>• Compressible and incompressible flow<br>• Viscous and inviscid flow<br>• Rotational and irrotational flow<br>• Comparison of flow types | L & Ds | PPT, PDF | 20 min |
| (c) | • Application of flow definitions<br>• Real-world examples of flow types<br>• Importance of ideal vs real fluids<br>• Summary of key distinctions<br>• Preparation for flow over bodies | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the difference between a streamline and a stream tube.
(b) Why is the concept of an ideal fluid useful in aerodynamics?
(c) How does turbulent flow differ from steady flow?

| | |
|---|---|
| **12. Home Assignment and Follow Up** | Prepare a brief note on how the concepts of streamline and stream tube will be applied when analyzing flow over bodies in the next lesson. |
| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Flow over bodies; bluff bodies and streamlined bodies, Flow pattern over flat plate, formation of eddies, the wake and its reduction. | Refer to the master syllabus reference list | LP 05 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Instructor, Flight Lieutenant, P No. 12345 | Senior Instructor, Captain, P No. 67890 |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min